

FIREHAWK TECHNICAL CHALLENGE

John's Car App

User guide for the live automobile database

Open the application: john-doe-automobile-database.web.app

At a glance Browse 398 automobile records, search and compare them, keep the last-used view, export CSV backups, and - with an owner account - add validated cars to the live database.

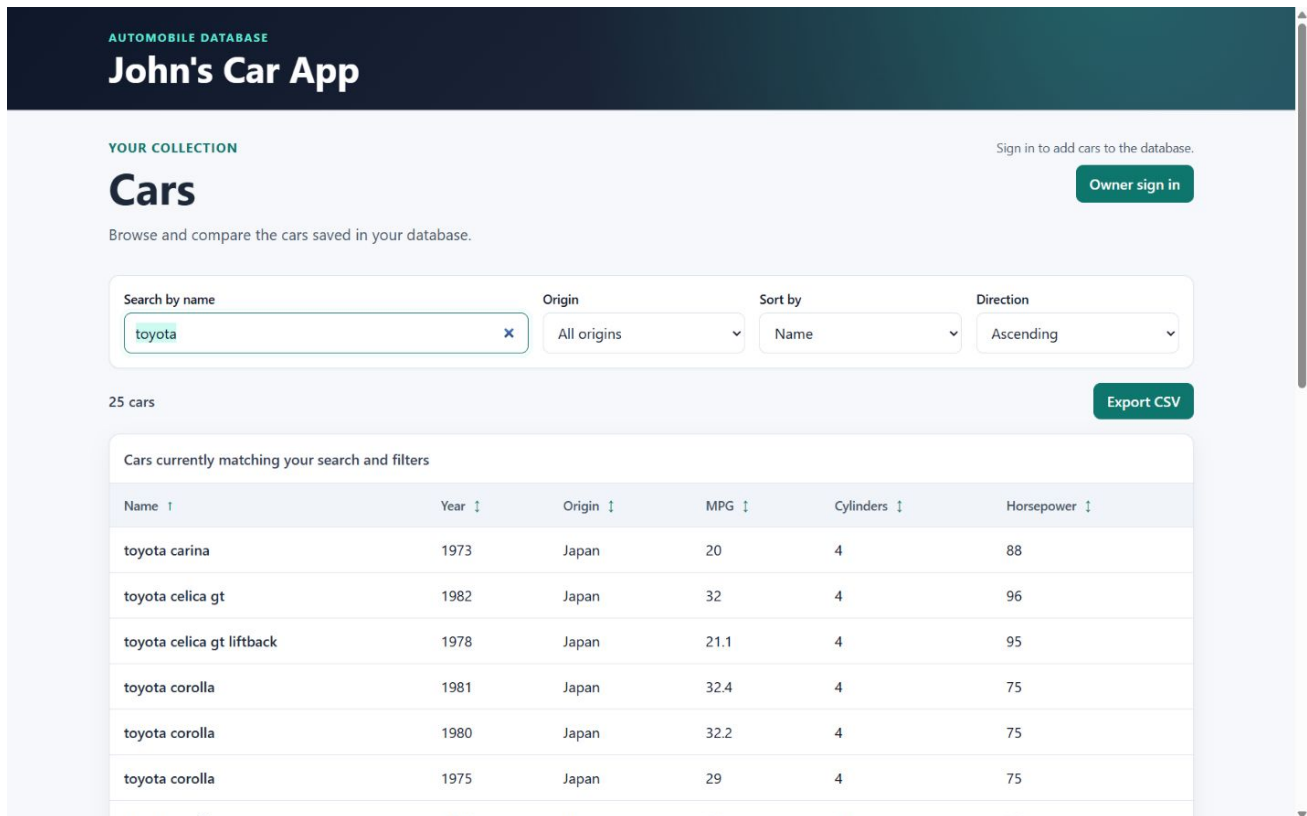


Figure 1. Live public Cars screen with owner sign-in and browsing controls.

Browse without an account

Everyone can view the database. Use the controls independently or together:

1. **Search by name** - enter all or part of a car name; the table updates immediately.
2. **Origin** - choose All origins, USA, Europe, Japan or Other.
3. **Sort** - choose Name, Year, Origin, MPG, Cylinders or Horsepower and a direction.
4. **Table headings** - click a heading to sort; the arrow shows the active direction.
5. **Export CSV** - download the currently matching records for Excel or backup.

Remembered view Search, origin, sort field and direction are saved in this browser and restored when the application is reopened.

Owner access

Browsing does not require an account. Creating cars is restricted to the authorised collection owner.

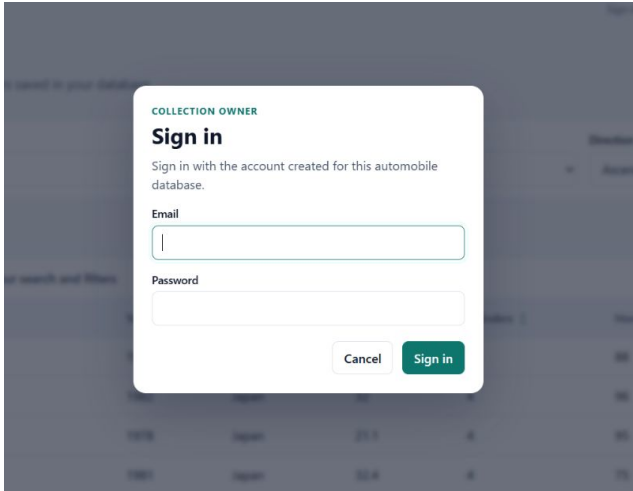


Figure 2. Secure owner sign-in.

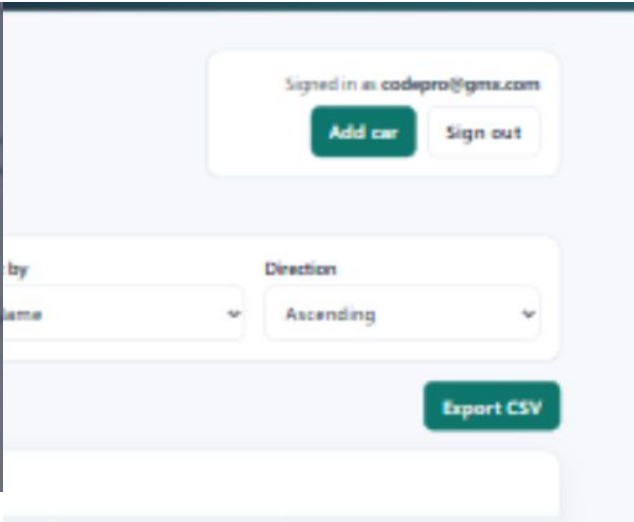


Figure 3. Authenticated owner controls.

Owner account	Access information
Email	codepro@gmx.com
Password	Obtain securely from the project owner. It is intentionally not printed in this guide.

Sign in step by step

1. Select **Owner sign in** at the top right of the Cars page.
2. Enter the owner email and the password supplied through a secure channel.
3. Select **Sign in**. The application shows the signed-in email, Add car and Sign out.
4. If sign-in fails, check the email and password, then try again. Do not share the password in email, source code, screenshots or public documents.

Security note The account can create production records. Sign out when finished. If a password is disclosed, ask the project owner to change it immediately.

Add a car

Select **Add car** to open the new database entry form. Required fields are marked with an asterisk; unknown optional measurements may be left blank.

Figure 4. The live Add a car form contains the complete automobile schema.

Complete the form

Field	Required?	What to enter
Name	Yes	A clear vehicle name, up to 120 characters.
Model year	Yes	A whole year from 1886 to 2100.
Origin	Yes	USA, Europe, Japan or Other.
Cylinders	Yes	A whole number from 1 to 16.
MPG	No	Fuel economy from 0 to 200.
Displacement	No	Engine displacement in cubic inches, 0 to 2000.
Horsepower	No	Horsepower from 0 to 5000.
Weight	No	Vehicle weight in pounds, 0 to 50000.
Acceleration	No	Acceleration time in seconds, 0 to 120.

Before selecting Add car Check spelling, units and the model year. The app compares all nine normalized fields when deciding whether the car is a duplicate.

Validation and duplicate protection

The form checks the entry before it writes to Firestore. Invalid fields are highlighted with a specific message and no car is created.

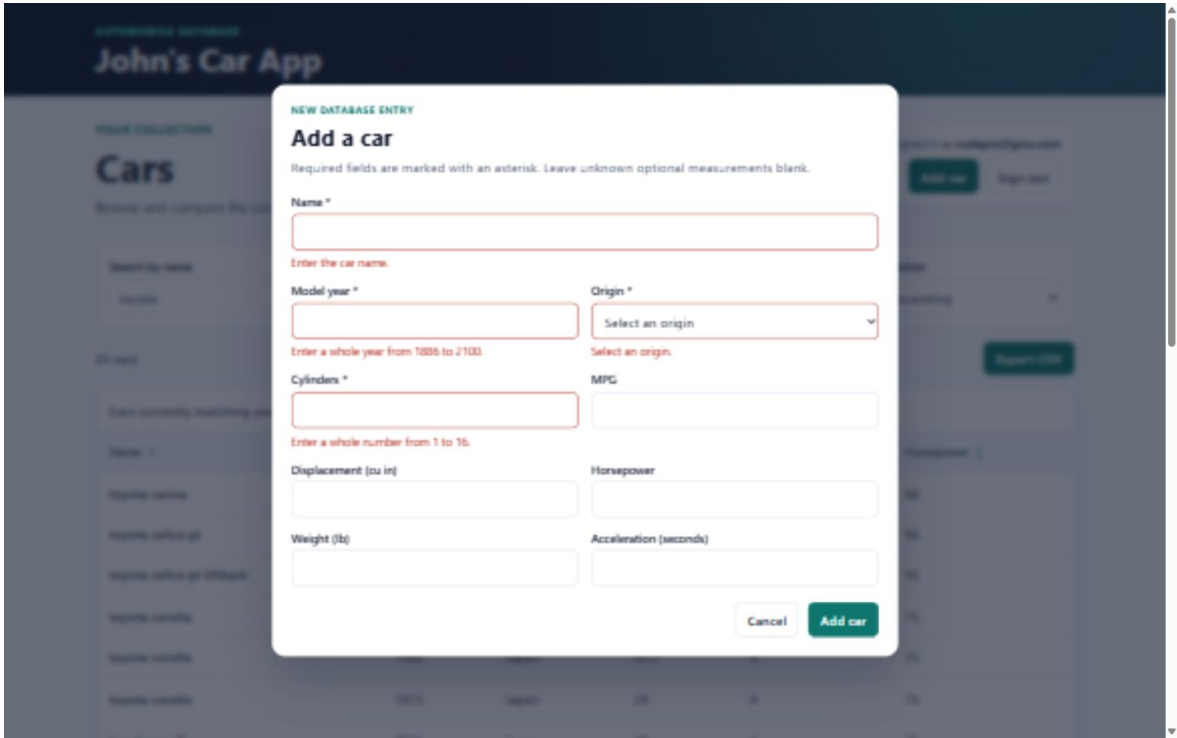


Figure 5. Required-field validation prevents an incomplete car from being submitted.

What happens after submission

Valid new car: the dialog closes, a success message appears, and the live table updates automatically.

Duplicate car: the app reports that the car already exists. The record count does not increase.

Connection or permission problem: the form remains available and shows an error. Check the connection and confirm the owner is still signed in before retrying.

Cancel: closes the dialog without saving. **Sign out:** removes the Add car control until the owner signs in again.

Duplicate definition Two entries are duplicates when their normalized name, MPG, cylinders, displacement, horsepower, weight, acceleration, model year and origin are all the same.

How the challenge requirements are met

Requirement / user need	App function
Simple interaction	One page, plain labels, live counts, readable tables and guided dialogs.
Search, filter and sort	Name search, origin filter, six sortable fields and clickable column headers.
Remember last view	Search, origin and sorting choices persist in the browser.
CSV backup	Export CSV downloads the currently matching dataset.
Future additions	An authorised owner can now add validated cars through the live interface.
Avoid duplicates	Normalized car data creates a deterministic identity; duplicates are rejected.
Firestore security	Public reads remain available; writes require the authorised owner account.

Dataset: Tawfik Elmetwally's Automobile Dataset (Kaggle). Live application checked 5 August 2026. This guide deliberately excludes passwords.